MILITARY STANDARD

MIL-STD-1325-183 (NAVY)

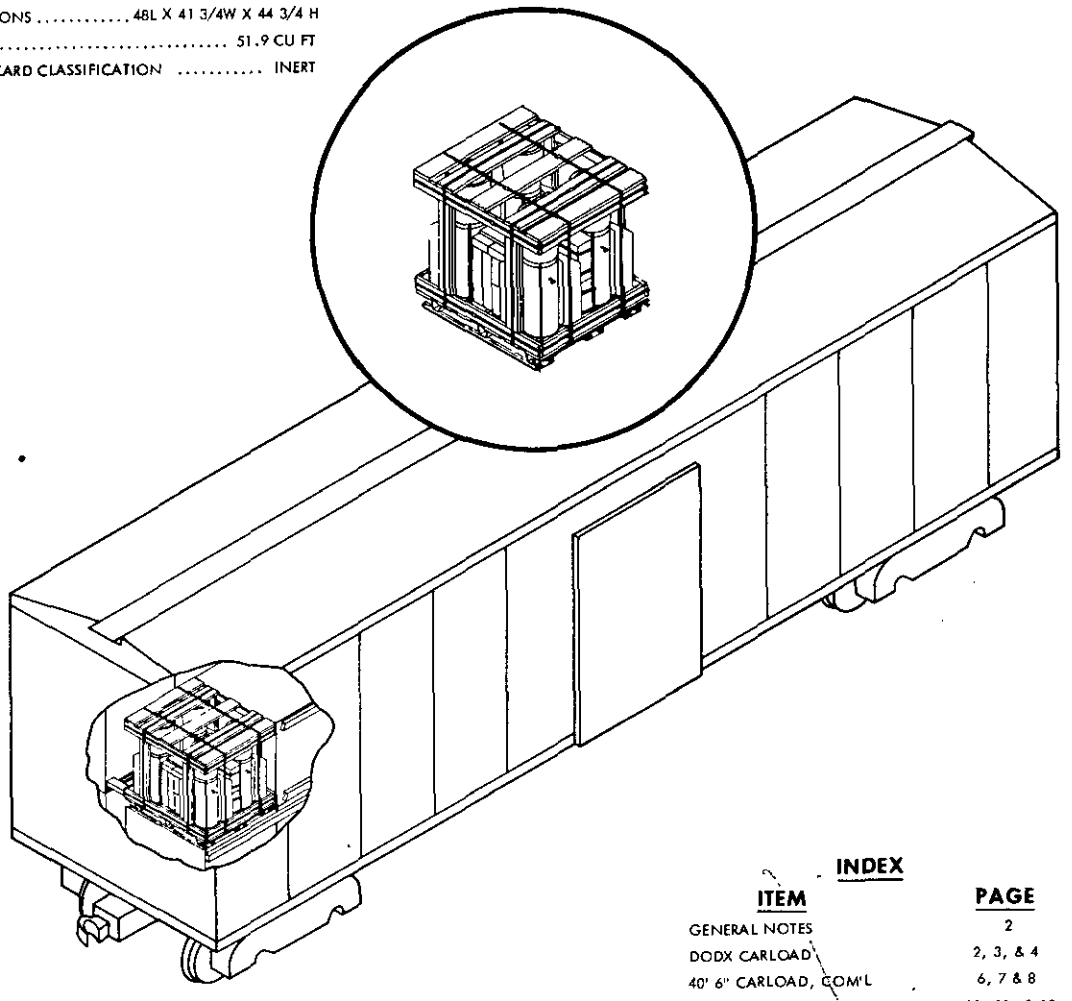
RAILCAR LOADING OF HAZARDOUS MATERIALS

9 AUGUST 1976

AIRFOIL GROUP, MXU-641/B, F/MK 83 BOMB
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/291
WEIGHT ..... 715 LBS
DIMENSIONS ..... 48L X 41 3/4W X 44 3/4 H
CUBE ..... 51.9 CU FT
DOT HAZARD CLASSIFICATION ..... INERT



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2, 3, & 4
40' 6" CARLOAD, COM'L	6, 7 & 8
50' 6" CARLOAD, COM'L	10, 11, & 12

NOTE:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINER-LOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M. S. Gray NWHC 8/6/76
TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 8/6/76
AIRSYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES NOT APPLICABLE

SIGNATURE SUPERVISOR, MILITARY & INTERMODEL SERVICES DATE

ORIGINATOR Charles McBride 8/6/76
DATE

NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 29000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF AIRFOIL GROUP MXU-641/B, F/MK 83 BOMB, PALLETIZED IN ACCORDANCE WITH WR-54/291.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORKLIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER - FED SPEC MM-L-751

    NAILS - FED SPEC FF-N-105

    STRAPPING - FED SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).

    SEALS - FED SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I).

50 FT 6 IN BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 52 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2300 LBS OF THE LADING.

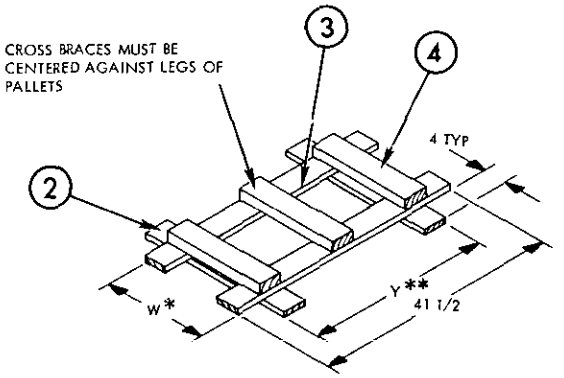
* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

** PIECE 3 NAILS TO PIECE 4 WITH 10D NAILS, 3 PER JOINT, CLINCHED, AND TO PIECE 2 WITH 6D NAILS, 1 PER JOINT

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
4	CROSS BRACE	2 X 4 X W*	39	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 X 4 X 41 1/2	26	2,4	SEE NOTE	**
2	SWAY BRACE SUPPORT	1 X 4 X (W+8)*	26	SEE 3	-	-
1	SLEEPER	2 X 4 X 40	26	CAR FLOOR	2 PER FOOT	16d

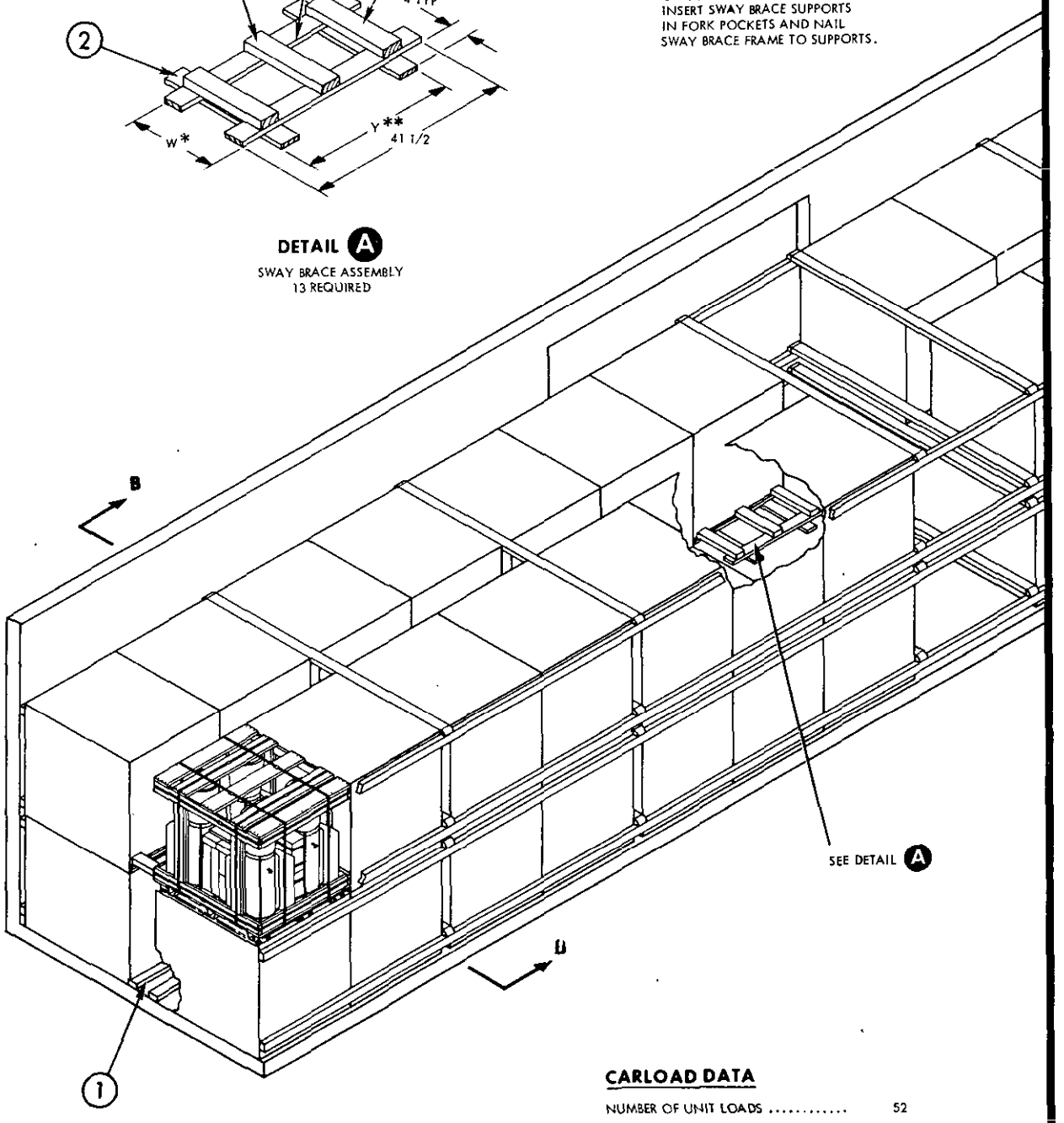
LIST OF MATERIALS & NAILING DATA

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* = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.
** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

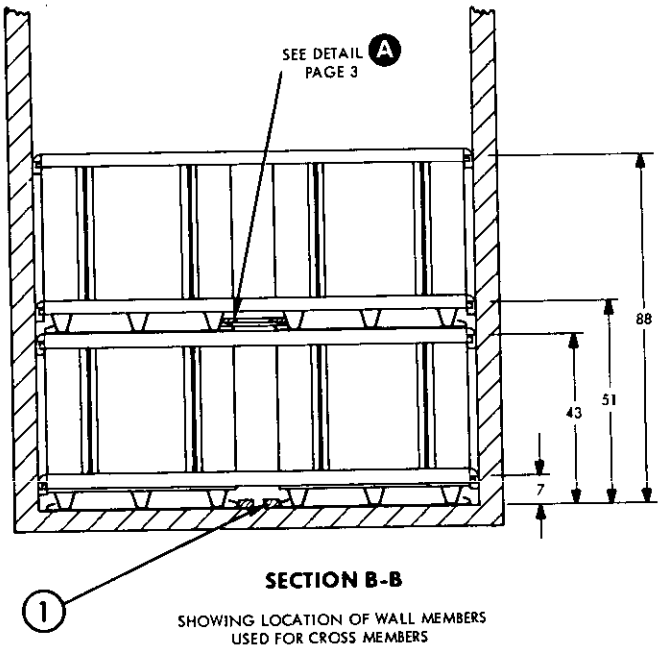
DETAIL A
SWAY BRACE ASSEMBLY
13 REQUIRED



CARLOAD DATA

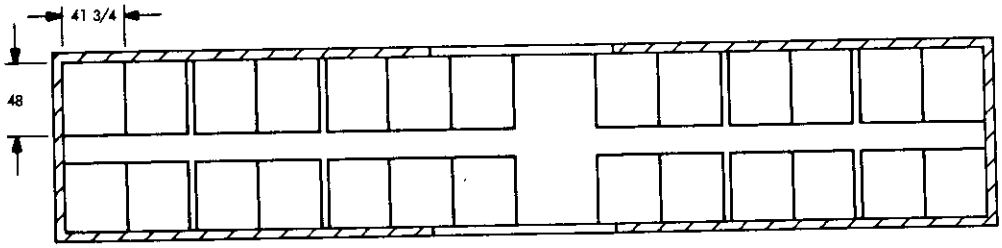
NUMBER OF UNIT LOADS	52
NUMBER OF CROSSMEMBERS REQ'D ...	24
LOAD WEIGHT (APPROX)	37180 LBS
DUNNAGE WEIGHT (APPROX)	338 LBS
CARLOAD WEIGHT (APPROX)	37518 LBS

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SECTION B-B

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSS MEMBERS



LOADING PLAN

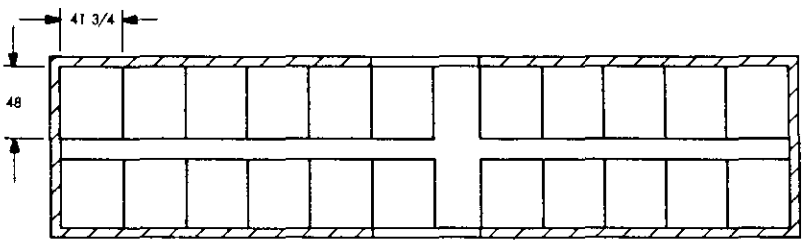
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40 FT 6 IN BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 44 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.



* USE SWAY BRACE ASSEMBLY DETAIL A FOR NON-NAILABLE FLOOR CARS.

** W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

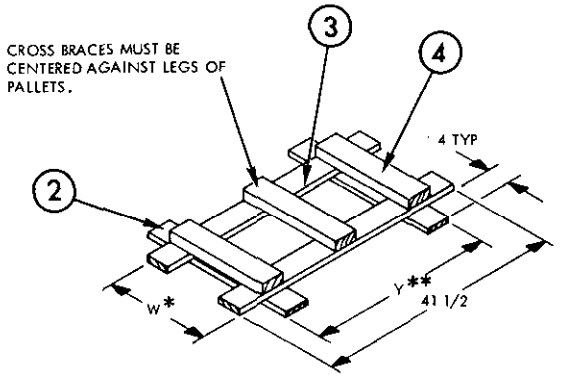
*** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

**** 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10D NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
13	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	8	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 X 3 X 96	4	DOOR POST	2 PER FOOT	20d
11	STRUT	4 X 4 X **** WEDGE FIT	16	6	2 PER JOINT	16d
10	HOLD DOWN	2 X 3 X TO SUIT	4	9	4	10d
9	HOLD DOWN	2 X 3 X TO SUIT	4	8	4	10d
8	HOLD DOWN SPACER	2 X 4 X CAR WIDTH-1	2	6	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 X 6 X 96	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 X 4 X W **	33	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 X 4 X 41 1/2	22	2, 4	SEE NOTE ***	
2	SWAY BRACE SUPPORT	1 X 4 X (W+8) **	22	SEE 3	-	-
1	SLEEPER *	2 X 4 X 40	22	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS & NAILING DATA

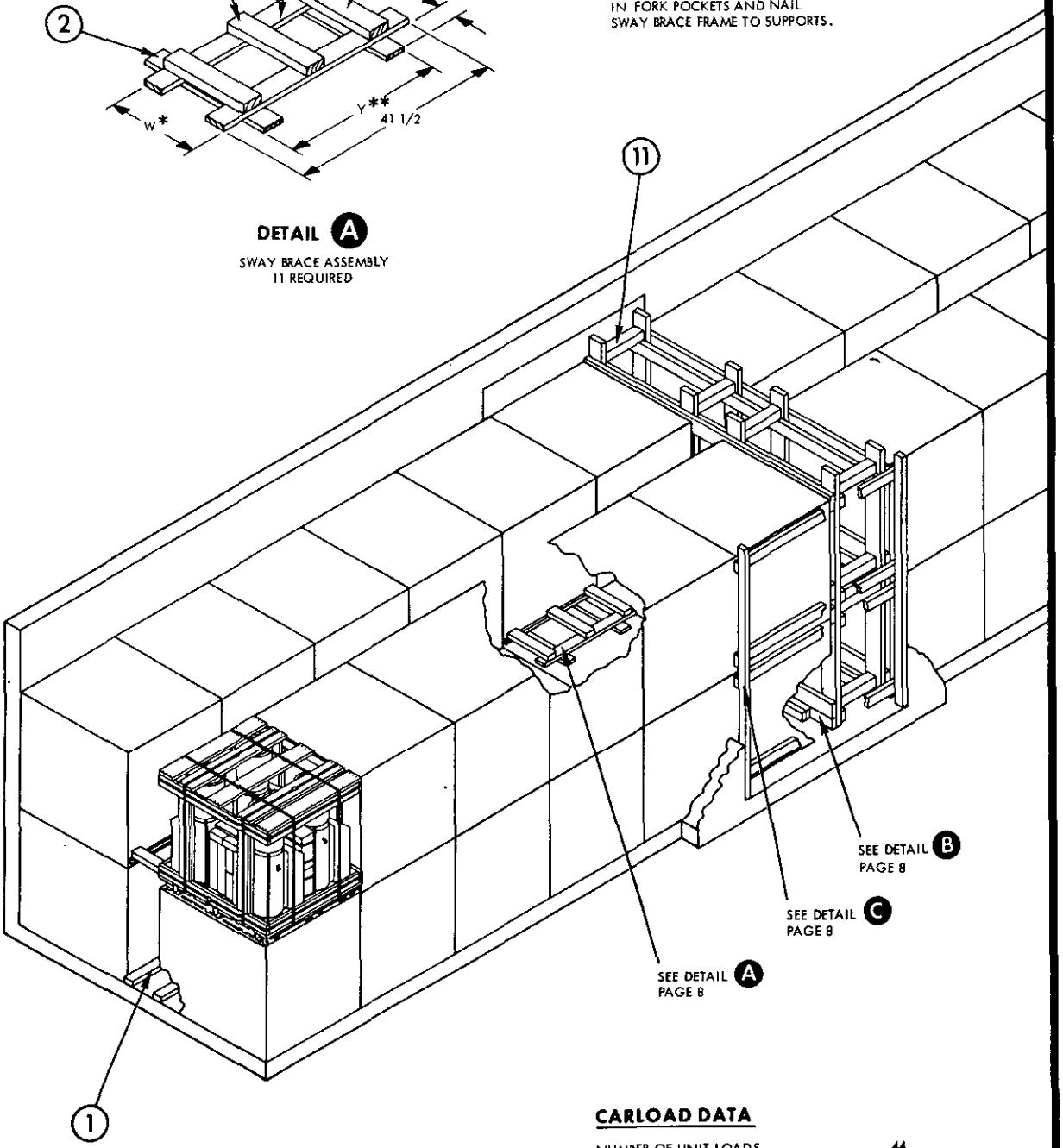
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* = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.
** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

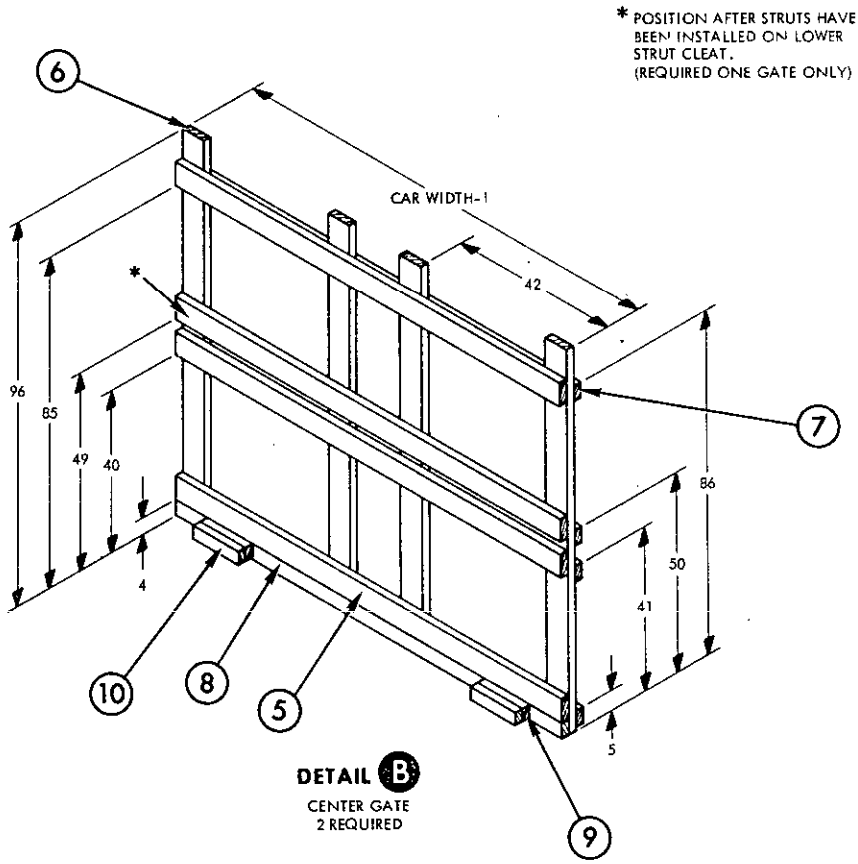
SWAY BRACE ASSEMBLY
11 REQUIRED



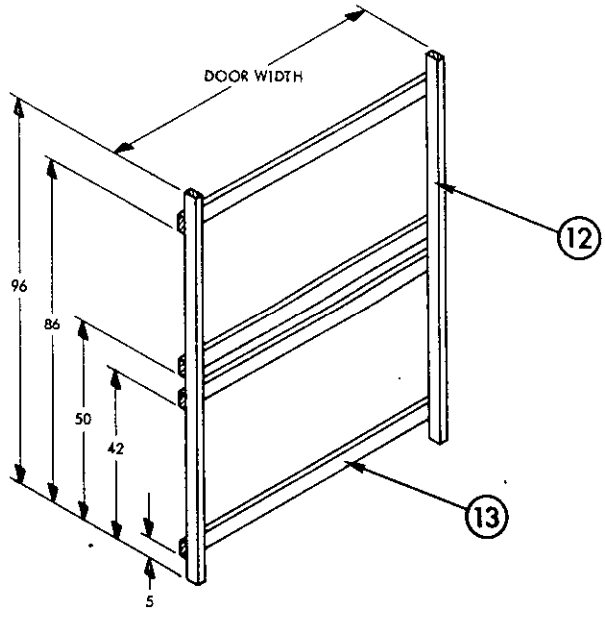
CARLOAD DATA

NUMBER OF UNIT LOADS	44
LOAD WEIGHT (APPROX)	31460 LBS
DUNNAGE WEIGHT (APPROX)	938 LBS
CARLOAD WEIGHT (APPROX)	32398 LBS

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DETAIL B
CENTER GATE
2 REQUIRED



WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED.

DETAIL C
DOORWAY PROTECTION
2 REQUIRED

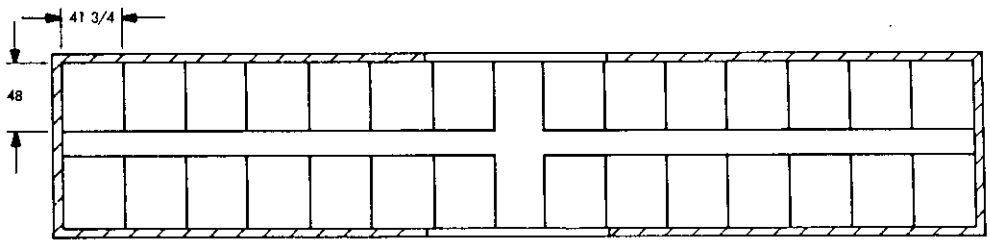
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50 FT 6 IN BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 56 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONEVNTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS, PROVIDING STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA ARE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE, CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL D) INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. EACH STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILLIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. DIMENSIONAL LUMBER DOORWAY PROTECTION, PIECES 12 AND 13, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DIMENSIONAL LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
13	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	8	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 X 3 X 96	4	DOOR POST	2 PER FOOT	20d
11	STRUT	4 X 4 X **** WEDGE FIT	16	6	2 PER JOINT	16d
10	HOLD DOWN	2 X 3 X TO SUIT	4	9	4	10d
9	HOLD DOWN	2 X 3 X TO SUIT	4	8	4	10d
8	HOLD DOWN SPACER	2 X 4 X CAR WIDTH-1	2	6	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 X 6 X 96	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 X 4 X W**	42	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 X 4 X 41 1/2	28	2, 4	SEE NOTE ***	
2	SWAY BRACE SUPPORT	1 X 4 X (W+8)**	28	SEE 3	-	-
1	SLEEPER*	2 X 4 X 40	28	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS & NAILING DATA

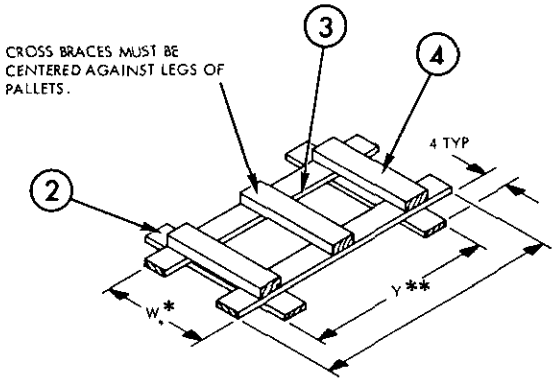
* USE SWAY BRACE ASSEMBLY DETAIL A FOR NON-NAILABLE FLOOR CARS

** W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

*** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

**** 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4's.

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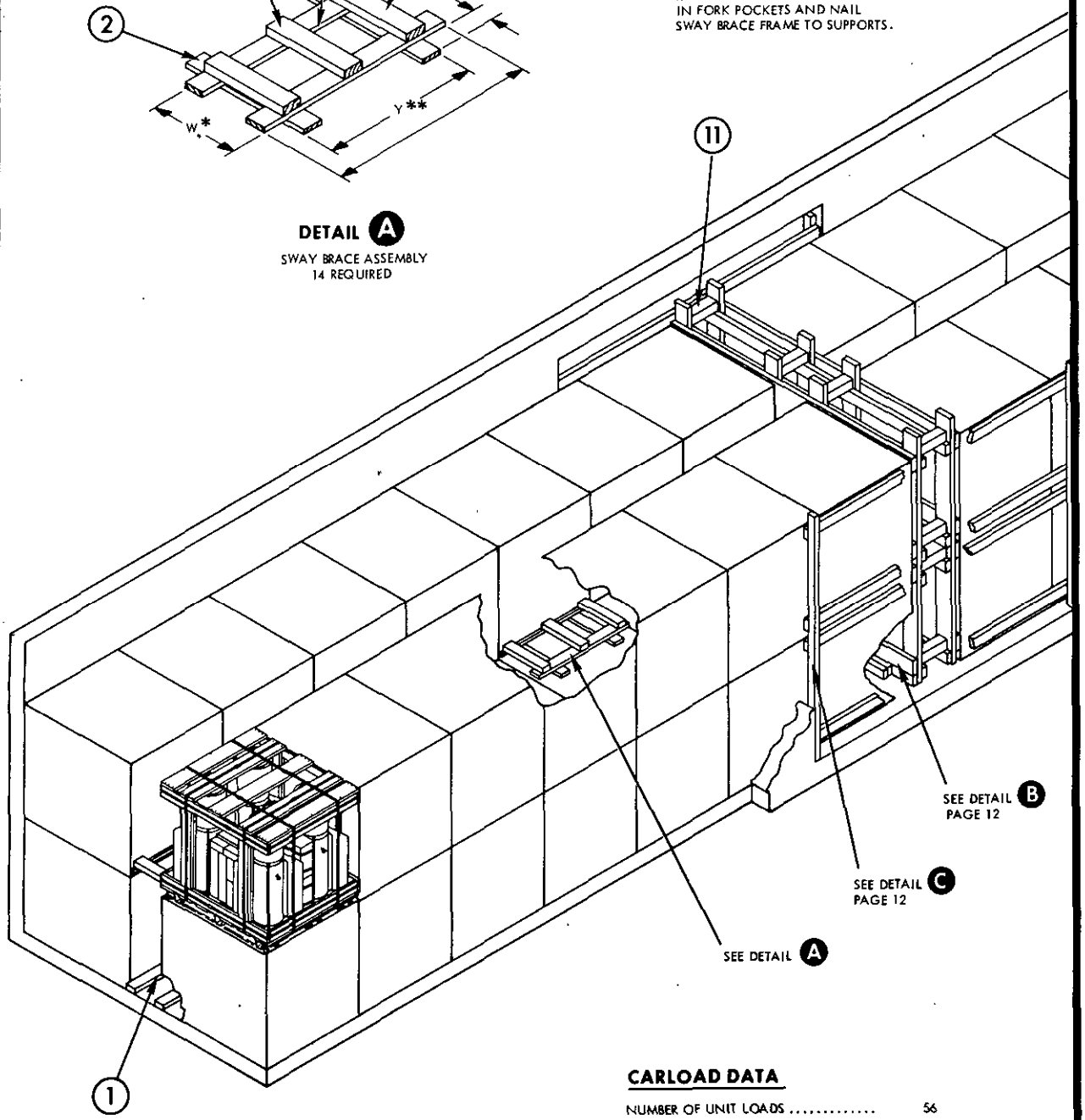


* = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** Y= DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

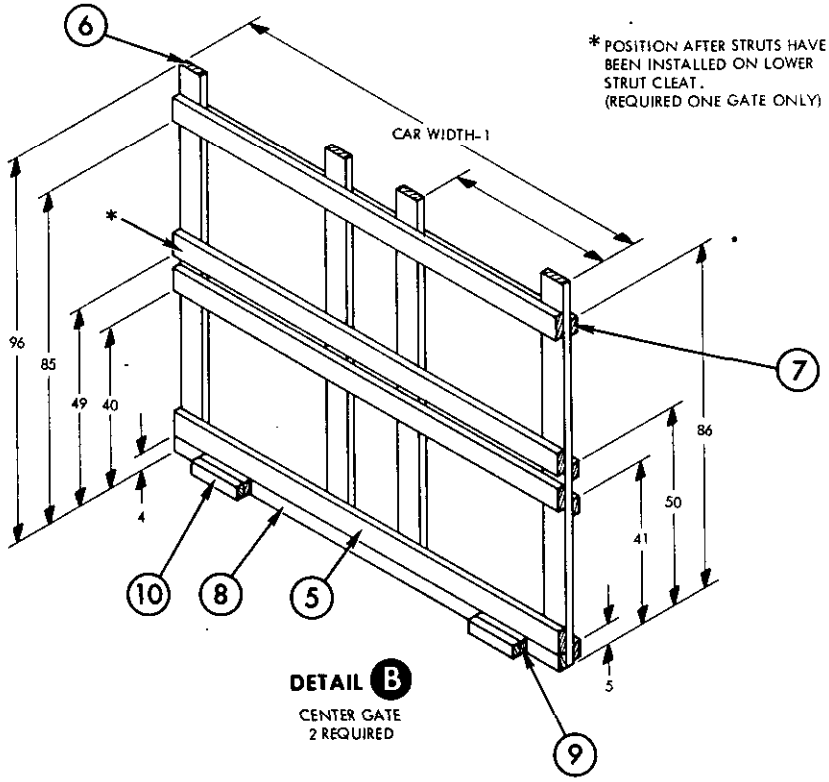
SWAY BRACE ASSEMBLY
14 REQUIRED



CARLOAD DATA

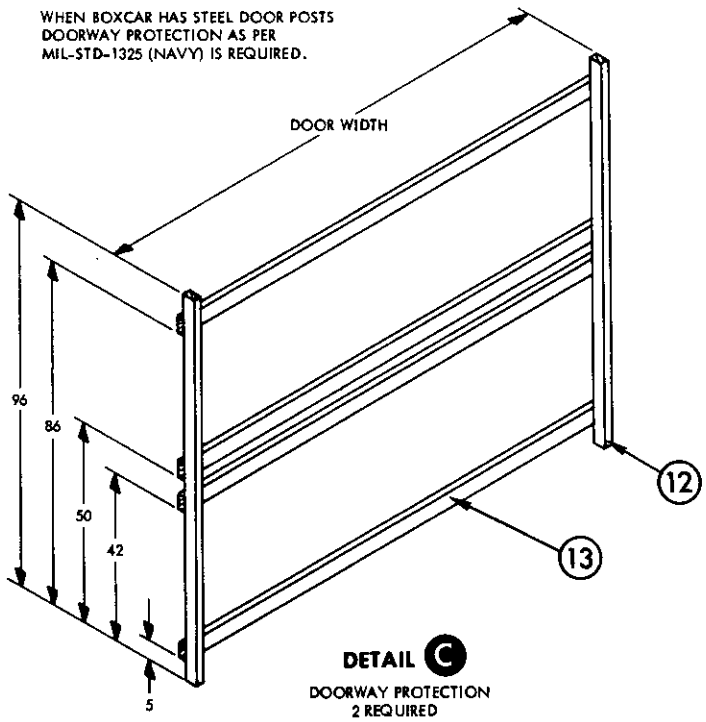
NUMBER OF UNIT LOADS	56
LOAD WEIGHT (APPROX)	40040 LBS
DUNNAGE WEIGHT (APPROX)	1044 LBS
CARLOAD WEIGHT (APPROX)	41084 LBS

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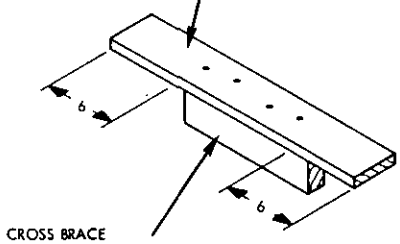
DETAIL B
CENTER GATE
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED.



DETAIL C
DOORWAY PROTECTION
2 REQUIRED

STRAPPING BOARD 1 X 4 X CUT TO FIT. NAIL TO EDGE OF CROSS BRACE WITH FOUR 10d NAILS.



CROSS BRACE 2 X 4 X CUT TO FIT SNUGLY BETWEEN LATERALLY ADJACENT PALLETS.

DETAIL D
TOP SWAY BRACE

REVIEW ACTIVITY
NAVY-OS, AS

PREPARING ACTIVITY
NAVY-OS
(PROJECT NO 8140-N341)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1325-183 (Navy)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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COMMANDING OFFICER
Naval Weapons Station Earle
Naval Weapons Handling Center (Code 803)
Colts Neck, New Jersey 07722

FOLD

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